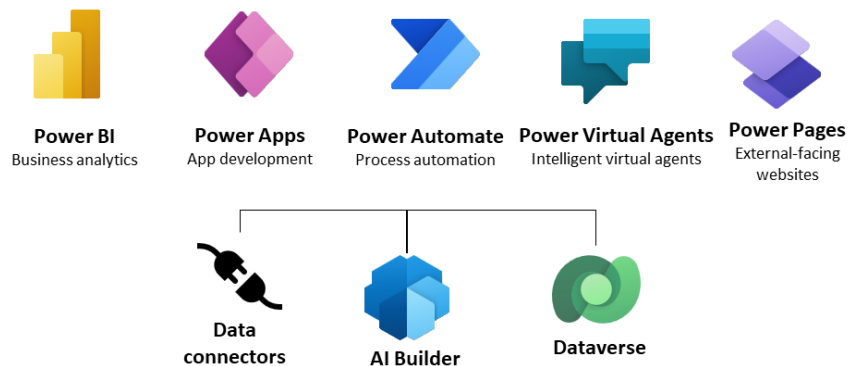

PL_200_Power_Platform_Functional_Consultant_202X

App making consists of the following areas:

- Modeling business data
- Defining business processes
- Composing the app
- Configuring security roles
- Sharing your app



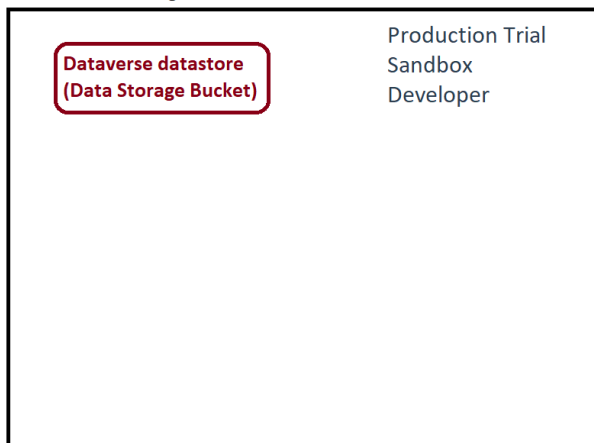
The low code platform that spans Microsoft 365, Azure, Dynamics 365, and standalone apps.



What & Where Can U - C.R.U.D.

<https://powerapps.microsoft.com>

Environment - Big Bucket GEO RBAC



Tools to Do the Work

Power Platform admin center
or with the Power Apps PowerShell cmdlets

Business Units Organizational

Business unit and child business unit

Only the user's own records

Contents

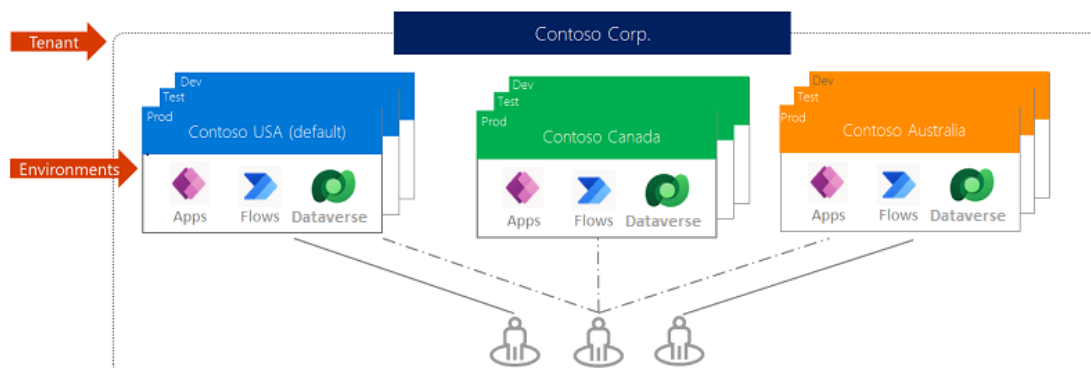
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The Functional Consultant

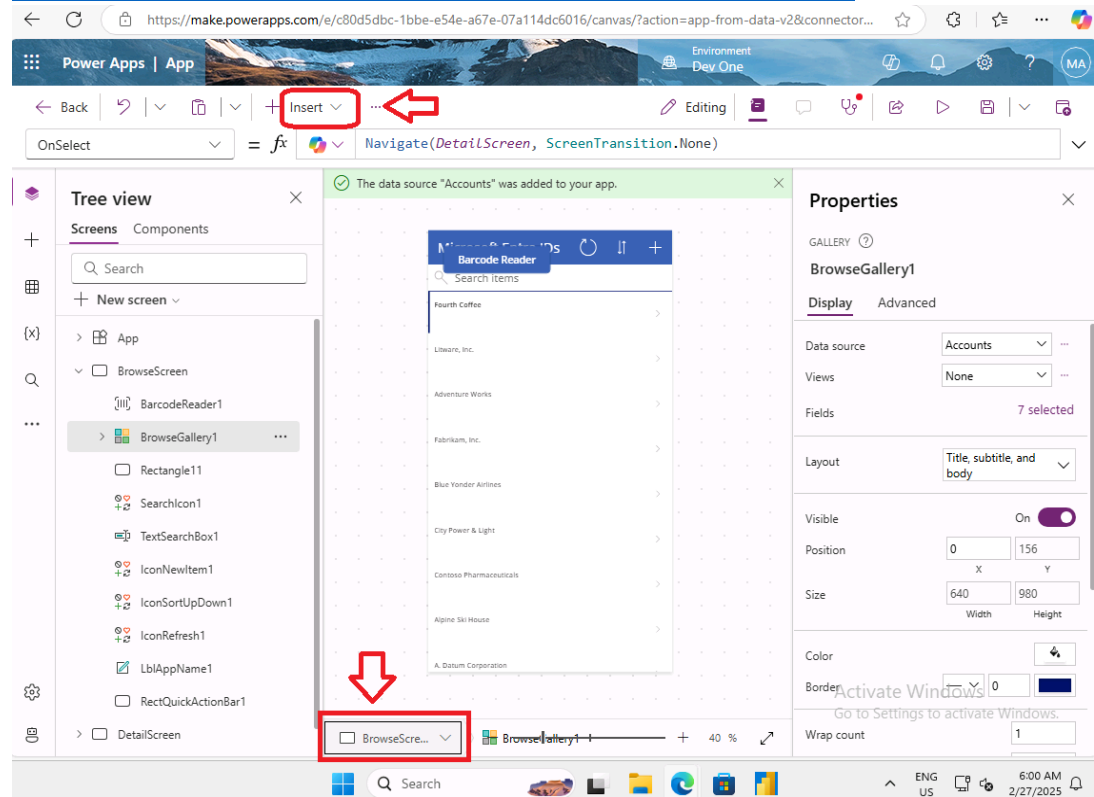
The Functional Consultant implements components of a solution that include application enhancements, tailored user experiences, process automation, and reporting.

* From MS Course Path

- [Course PL-200T00--A: Microsoft Power Platform Functional Consultant - Training | Microsoft Learn](#)
- [Exam PL-200: Microsoft Power Platform Functional Consultant - Certifications | Microsoft Learn](#)
- [GitHub - MicrosoftLearning/PL-200-Power-Platform-Functional-Consultant: PL-200 Power Platform Functional Consultant](#)
- <https://aka.ms/ppac> URL in labs to get to portal
 - Resolves to <https://admin.powerplatform.microsoft.com/>
- [Data requirements - Training | Microsoft Learn](#)
- [Dataflows - Training | Microsoft Learn](#)
 - Dataflows are cloud-based extract, transform, and load (ETL) services that help you ingest data from various sources and then convert it into an analysis-ready form.
- [What is Power Apps? - Power Apps | Microsoft Learn](#)
 - [Understand Power Apps Studio - Power Apps | Microsoft Learn](#)
 - The Editor
 - [Overview of building canvas apps - Power Apps | Microsoft Learn](#)
 - Design and build a business app from a canvas in Microsoft Power Apps without writing code by dragging and dropping elements onto a canvas,
 - Create Excel-like expressions for specifying logic and working with data.
 - Integrate business data
 - Share your app - browser - mobile device
 - Embed in SharePoint, Power BI, Teams or model-driven app for mobile or tablet devices, where you'll need task-based applications.
 - ModelDrivenFormIntegration control properties and actions
 - This control is responsible for bringing contextual data from the host model-driven app to the canvas app
 - endlist



- [Overview of building a model-driven app with Power Apps - Power Apps | Microsoft Learn](#) (Looks Like Power BI for Dataverse)
 - Without a data model housed within Microsoft Dataverse, you can't create a model-driven app.
 - Process driven apps that are data dense - make it easy for users to move between related records.
 - Forms, views, and charts and dashboards to tables using an app designer tool.
 - Relationships connect tables together in a way that permits navigation between them and ensures that data is not repeated unnecessarily.
 - [App navigation in model-driven apps - Power Apps | Microsoft Learn](#)
 - Area – Group – Page (Subarea that contains data)
 - Page – Tables, dashboards, URLs – Custom
 - Data is stored in tables in Microsoft Dataverse, and forms provide the user interface (UI) for interacting with the data. Controls on the form allow users to visualize data that is in table columns.
- [Controls and properties in canvas apps - Power Apps | Microsoft Learn](#)



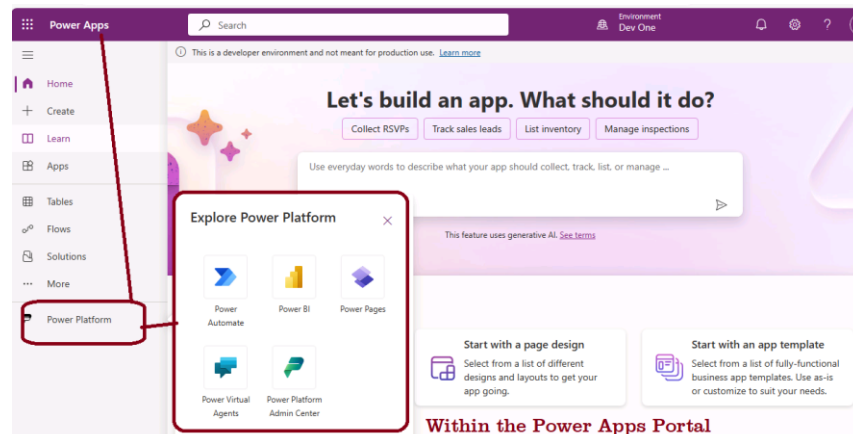
- Space

Dataverse

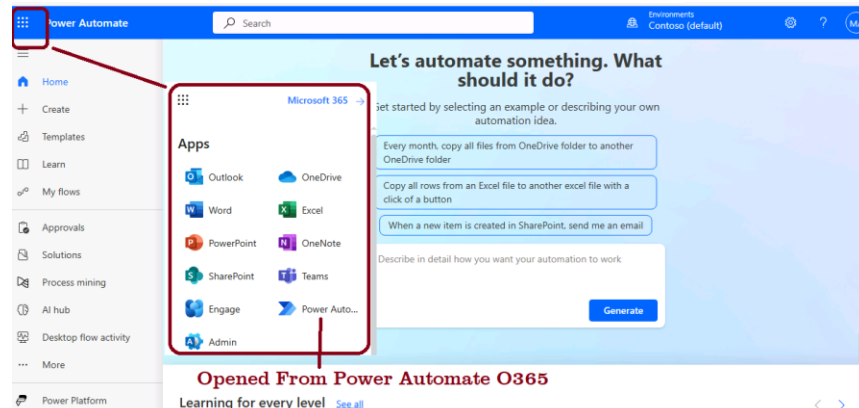
- [What is Microsoft Dataverse? - Power Apps | Microsoft Learn](#)
 - Dataverse allows data to be integrated from multiple sources into a single store, which can then be used in Power Apps, Power Automate, Power BI, and Power Virtual Agents along with data that's already available from the Dynamics 365 applications.
 - Table is a set of rows (formerly referred to as records)
 - [View data in a table - Training | Microsoft Learn](#)
 - Columns (formerly referred to as fields/attributes).
 - Environment Maker and Environment Admin are the only predefined roles for environments that have no Dataverse database.
 - As a Dataverse database, a user must be assigned the System Administrator role instead of the Environment Admin role to have full admin privileges.
 - Terminology updates
 - Virtual entities and dual-write are part of the shared data layer for the convergence of finance and operations apps and the Dataverse platform.
 - Dual-write is your best option when your data resides completely inside Dataverse
 - Virtual entities seamlessly represent external data inside of Dataverse without replicating the data.
 - Every table contains a Primary name column that is used to reference the table row within the user interface.
 - Primary key column is a GUID.
 - Column types
 - [Define rollup columns with Power Apps - Power Apps | Microsoft Learn](#)
 - Choice columns provide a set of options that are displayed in a drop-down control on a form or in picklist control when you're using Advanced Find.
 - Autonumber columns offer many options for the type of autogenerated number that you want to generate.
 - [Relate one or more tables - Introduction - Training | Microsoft Learn](#)
 - An easy way to create a table relationship is by creating a column with data type Lookup to another table.
 - Creates a many-to-one relationship.
 - - Correspondingly, creating a one-to-many relationship creates a lookup column on the related table.
 - [*Exercise - Create two tables and relate them by using a one-to-many relationship - Training | Microsoft Learn](#)
 - [*Exercise - Create a cloud flow with a Dataverse connector - Training | Microsoft Learn](#)
 - FetchXML is a proprietary XML-based query language of Dataverse that you can use to query data.
 -
- [Demo Videos and Guided Tours | Microsoft Power Apps -](#)

Portals & Tools

1. Power Platform admin center
 - a. admin.powerplatform.microsoft.com
 - b. Create and manage environments
 - c. View Dataverse analytics
 - d. Real-time, self-help recommendations.
2. Power Apps portal
 - a. <https://make.powerapps.com/>
3. Developer
 - a. [Dataverse developer documentation - Power Apps | Microsoft Learn](#)
 - b. [Model-driven apps developer documentation - Power Apps | Microsoft Learn](#)
 - c. [Canvas apps for enterprise developers, partners, and ISVs - Power Apps | Microsoft Learn](#)
4. Power Automate is an online workflow service that automates actions across the most common apps and services.
 - a. Every flow has two main parts: a trigger, and one or more actions.
 - b. <https://make.powerautomate.com/>



c.



d.

5. Power Pages portal
 - a. <https://make.powerpages.microsoft.com/>
6. App designer
7. space

2024 & Changes Reference

- [*Introducing Power Automate - Training | Microsoft Learn](#)[Get started with Power Automate - Power Automate | Microsoft Learn](#)
- [*Exercise - Create recurring flows - Training | Microsoft Learn](#)
- [Create and manage flows in Power Apps. - Power Automate | Microsoft Learn](#)
- [Get started with Copilot in cloud flows - Power Automate | Microsoft Learn](#)
- [Get contextual help with flows from the Microsoft Copilot Studio bot - Power Automate | Microsoft Learn](#)
- [Mobile offline overview - Power Apps | Microsoft Learn](#)
 - Offline-first means that all the data you might need when offline is copied to your mobile device
 - Sync data between your mobile device and Microsoft Dataverse
 - Offline-first vs. classic offline
- Space

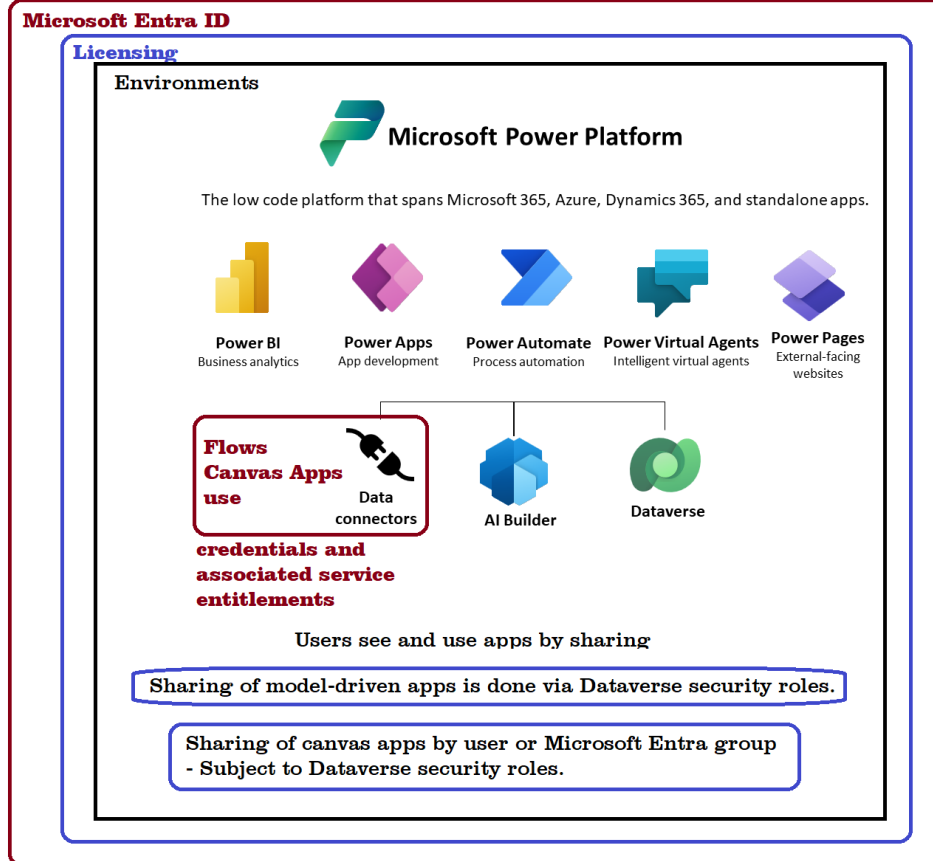
RBAC & Roles

A user's access to a record is the combination of all their security roles, the business unit they're associated with, the teams they're members of and the records that are shared with them.

1. [Security in Microsoft Dataverse - Power Platform | Microsoft Learn](#)
 - a. Combine business units, role-based security, row-based security, and column-based security
 - b. Users are authenticated by Microsoft Entra ID.
 - c. Licensing is the first control-gate to allowing access to Power Apps components.
 - d. [Manage Microsoft Dataverse settings - Power Platform | Microsoft Learn](#)
 - i. Ability to create applications and flows is controlled by security roles in the context of Environments.
 - e. A user's ability to see and use apps is controlled by sharing the application
2. [Security concepts in Microsoft Dataverse - Power Platform | Microsoft Learn](#)
 - a. Business Unit – for grouping users.
 - b. Teams are owned by a Business Unit.
 - c. Every Business Unit has one default team that is automatically created when the Business Unit is created.
3. [Configure user security in an environment - Power Platform | Microsoft Learn](#)
 - a. Summary of resources available to predefined security roles
4. [Configure user security in an environment - Power Platform | Microsoft Learn](#)
 - a. Create a custom security role with minimum privileges to run an app
5. [Create users - Power Platform | Microsoft Learn](#)
6. [Control user access to environments: security groups and licenses - Power Platform | Microsoft Learn](#)
 - a. Dataverse group team
 - b. Create a security group and add members to the security group
 - c. Create a user and assign license
 - d. Associate a security group with an environment

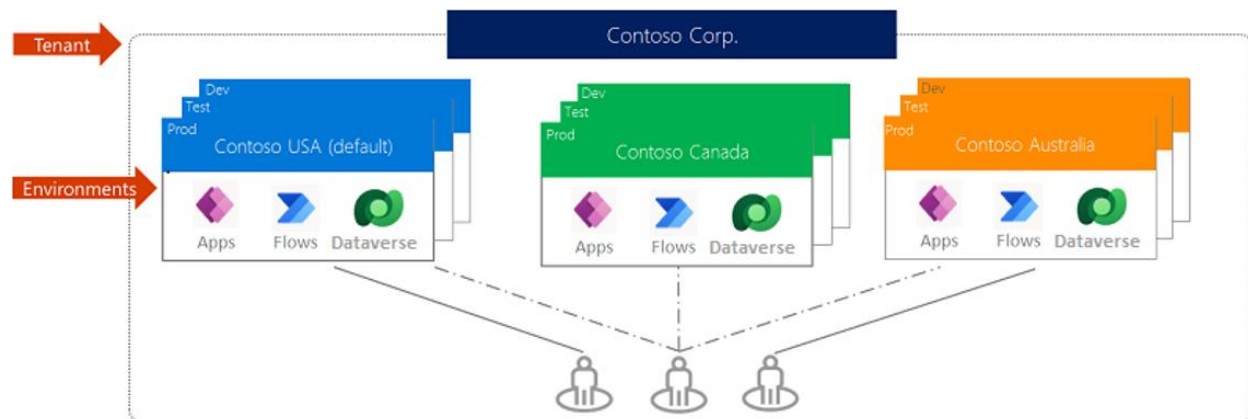
e. [Managed Environments overview - Power Platform | Microsoft Learn](#)

Security Boundaries



- 7.
8. A key concept of Dataverse security to understand is that privileges are accumulative with the greatest amount of access prevailing.
9. space

Parts



Power BI, Power Apps, Power Automate, Power Virtual Agents, and Dynamics 365 applications can get the data produced by the dataflow by connecting to Dataverse, a Power Platform dataflow connector.

- **Tables:** Allows you to manage the Microsoft Dataverse tables deployed in this environment. You can easily create new tables and perform tasks such as modifying the different forms, views, and relationships in the Dataverse instance.
- **Connections:** Allows you to manage the connections being used by apps in this environment.
- **Flows:** Provides access to any flows created for this environment.
- **Chatbots:** Provides access to any chatbots created for this environment.
- **AI Models:** Provides access to AI builder models for this environment.
- **Solutions:** Provides access to any Solutions deployed to this environment.
- **Cards:** Provides access to Cards created in this environment.
- **Choices:** Allows you to manage the choice columns in this environment.
- **Connections:** Allows you to manage the connectors used in this environment.
- **Dataflows:** Provides access to Dataflows used in this environment.
- **Power Platform:** Provides access to other Power Platform Maker portals.

[Managed Environments overview - Power Platform | Microsoft Learn](#) a suite of premium level capabilities admins can use to manage Power Platform at scale.

1. [Cards for Power Apps overview - Power Apps | Microsoft Learn](#)
 - a. Card forms are used in views for model-driven apps.
 - b. Present information
 - c. Added to views by using the Read-only Grid control.
2. [Tutorial: Customize a gallery in Power Apps - Power Apps | Microsoft Learn](#)
 - a. A list of records is called a gallery
3. [Customize forms in canvas apps - Power Apps | Microsoft Learn](#)
 - a. Each form comprises one or more cards, each of which shows data from a particular column in the data source.
 - b. C.R.U.D. Ops
 - i. Display form control
 - ii. Edit form control

4. Main forms - are the primary user interface where people view and interact with their data.
5. Quick Create - Model-driven apps and Dynamics 365 mobile apps use quick create forms for creating new rows. Space.
6. Quick view form can be added to another form as a quick view control.
7. space

Power Apps, Power Pages & Canvas Apps

Power Pages and **Canvas Apps** are both specialized product types *under* the broader **Power Apps** umbrella framework. Microsoft positions Power Apps as the overarching suite, meaning any conversation about the crossover between them is a comparison of specialized tools within the same application platform [1]. [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#)

The visual table below details how these development tools cross over, share components, and differ based on target audience and deployment needs:

Architectural & Functional Breakdown

Feature / Capability [1] , [2] , [3] , [4] , [5]	Power Pages	Canvas Apps	Power Apps (Overarching Suite)
Core Purpose	Public-facing websites and secure external portals.	Tailored, pixel-perfect internal task and mobile apps [1, 4].	The complete ecosystem enclosing all app/portal types [1].
Target User Base	External users (customers, partners) and internal staff.	Internal employees within your Microsoft 365 tenant [1, 4].	Scalable to both internal enterprise users and external guests [1].
Data Engine (Crossover)	Shared Dataverse integration for tables, forms, and views [5].	Connects to Dataverse , SharePoint, SQL, and 1,000+ APIs [1].	Standardizes on Dataverse as the native cloud data layer [1].
Design Framework	Responsive Bootstrap-based web grid system [5].	Absolute positioning (Drag-and-drop canvas canvas) [1, 4].	Supports web grid systems, canvas layouts, and model-driven UI [1].
Authentication System	External providers (Azure AD B2C, Google, Facebook, Okta).	Native Microsoft 365 / Entra ID enterprise login [1, 4].	Handles both enterprise IAM and custom external federated identity [1].

Core Areas of Crossover

- **The Dataverse Connection:** This is the strongest point of crossover. A table created in Microsoft Dataverse can simultaneously drive the data engine behind an internal **Canvas App** for your

employees and populate a public-facing data form on a **Power Pages** website for your vendors [5]. [1]

- **Power Automate Extensibility:** Both platforms natively use [Power Automate cloud flows](#) to run backend automation, process payments, send email alerts, or transfer files when a user interacts with the UI. [1, 2, 3, 4, 5]
- **Component Sharing (PCF):** Professional developers can build custom user interface elements using the **Power Apps Component Framework (PCF)**. Once built, these identical components can be deployed inside both Canvas Apps and Power Pages sites. [1, 2]
- **Unified Lifecycle (ALM):** Both app types are packaged into Microsoft Power Platform **Solutions**. This allows IT administrators to move a bundled package containing your Canvas Apps, Power Pages configurations, and cloud flows across dev, test, and production environments together.

When to Use Which?

- Choose [Power Pages](#) if you need to build a secure, authenticated website where people *outside* your company corporation (like citizens, patients, or contractors) can log in and submit data. [1, 2, 3, 4, 5]
- Choose [Canvas Apps](#) if your target audience is internal, and you need a highly specific mobile layout, offline capabilities, or a tool that talks to legacy sources like SharePoint lists

Quick Starts

1. [*Exercise - Create a custom table and import data - Training | Microsoft Learn](#)
2. [*Exercise - Training | Microsoft Learn](#)
 - a. Create a model-driven app
3. [*Form elements - Training | Microsoft Learn](#)
 - a. Form order defines the order in which a user sees the available forms, within the set of allowed forms for their security roles.
 - b. You can assign a security role (or collection of security roles) to control access to the form.
4. [*Exercise - Create your first app in Power Apps - Training | Microsoft Learn](#)
5. [Build your first modern model-driven app - Power Apps | Microsoft Learn](#)
6. [Design model-driven apps by using the app designer - Power Apps | Microsoft Learn](#)
7. [Create a canvas app with data from Microsoft Dataverse \(contains video\) - Power Apps | Microsoft Learn](#)
8. [Create a canvas app that can trigger a Power Automate flow \(contains video\) - Power Apps | Microsoft Learn](#)
9. [Add data to a table in Microsoft Dataverse by using Power Query - Power Query | Microsoft Learn](#)
10. [Manage behavior settings - Power Platform | Microsoft Learn](#)
11. [Create a machine ordering app with Power Apps - Online Workshop - Training | Microsoft Learn](#)
– another lab to build

Integrating Power Apps

1. [Scenarios for integrating SharePoint with Power Apps - Power Apps | Microsoft Learn](#)
2. space

Areas

1. [Get started using Microsoft Dataverse – Training | Microsoft Learn](#)
2. [Create and edit tables using Power Apps - Power Apps | Microsoft Learn](#)
3. [Create a relationship between tables in Microsoft Dataverse - Training | Microsoft Learn](#)
4. [Create and define calculation or rollup columns in Microsoft Dataverse - Training | Microsoft Learn](#)
5. [Load/export data and create data views in Microsoft Dataverse - Training | Microsoft Learn](#)
6. [Manage permissions and administration for Microsoft Dataverse - Training | Microsoft Learn](#)
- 7.

General Notes

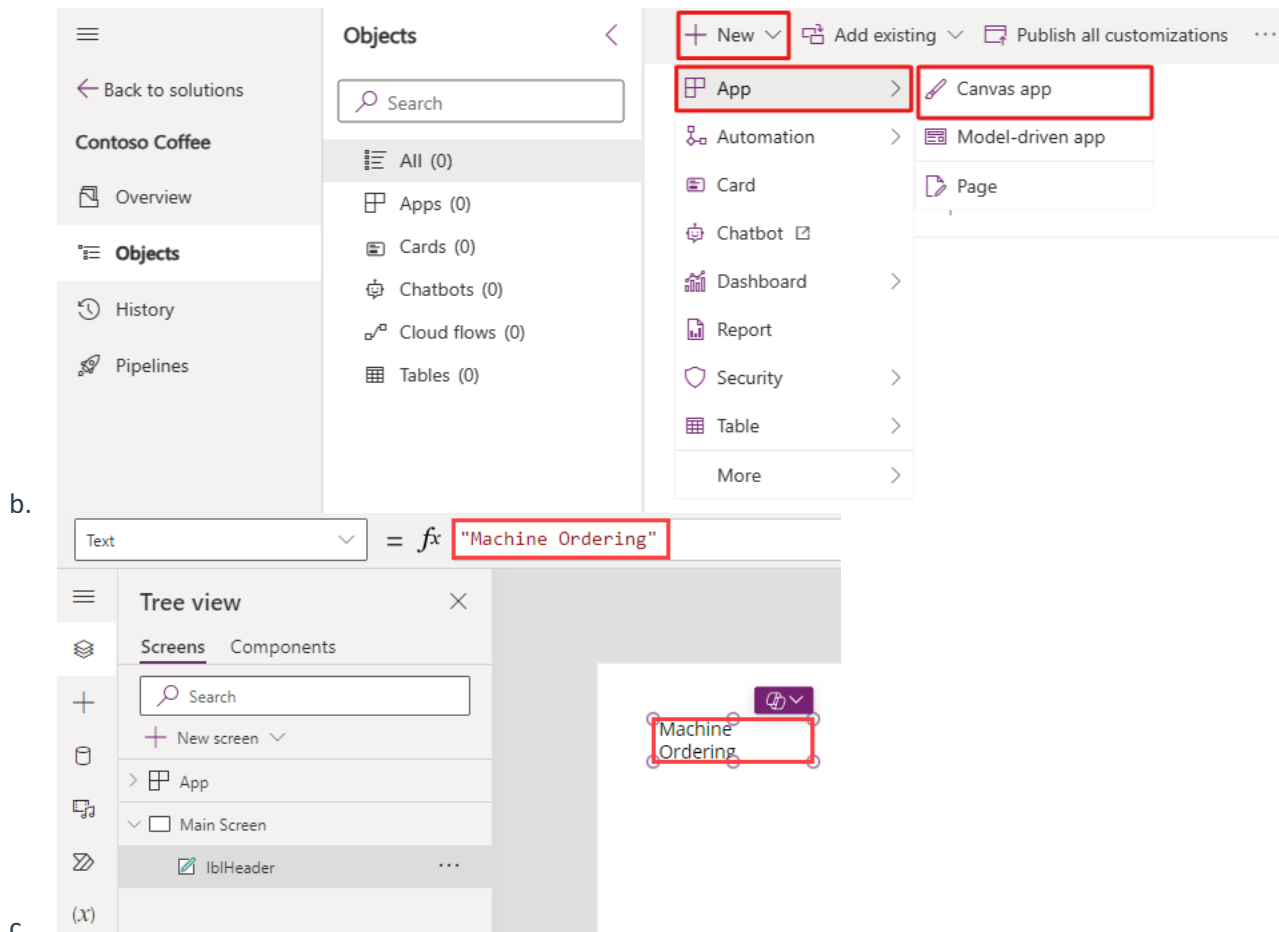
Portals to Automate Stuff

These need subscription to use:

- Power Apps maker portal <https://make.powerapps.com>
- Power Platform admin center <https://admin.powerplatform.microsoft.com>

Things to Remember

1. Dataverse there's also a column called the primary column
2. A virtual table is a custom table in Microsoft Dataverse that has columns containing data from an external data source, such as Azure SQL Database or SharePoint.
3. Schema name: This is the internal name of the column created in the database. By default, the schema name is automatically created for you based on the Display name.
4. To hide a standard table, change the security role privileges for your organization to remove the Read privilege for that table.
5. Power Apps web studio
 - a. Environment - Solution – Publisher – App –

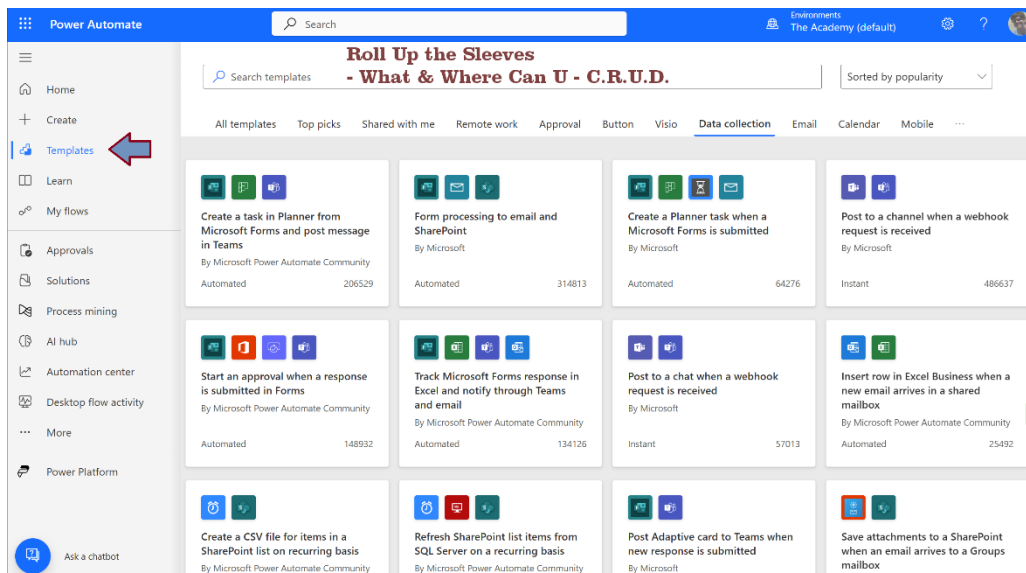
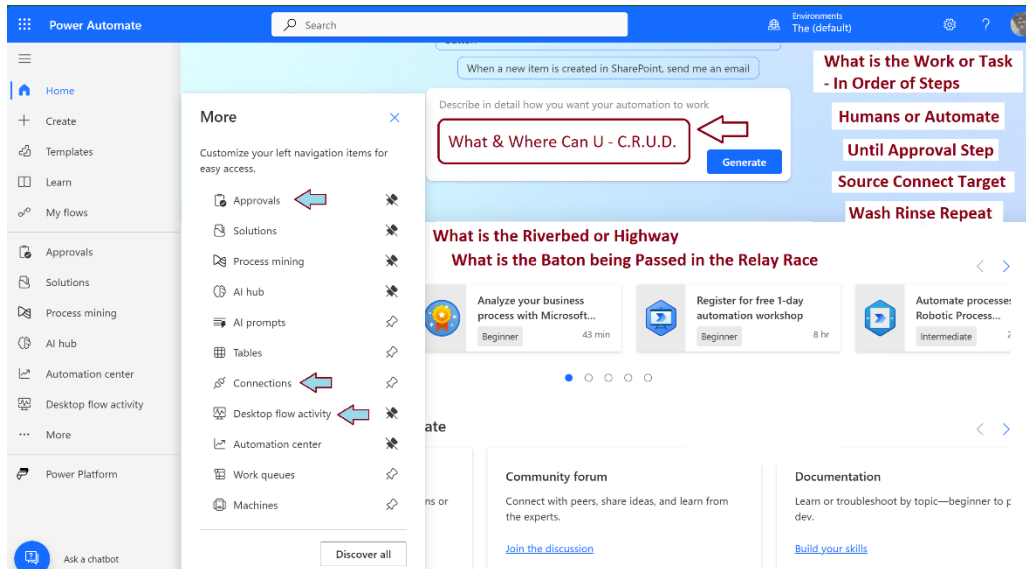


- c.
- d. [Use Power Fx - Training | Microsoft Learn](#)
- e. DAX
- f. [Microsoft Power Platform CLI - Power Platform | Microsoft Learn](#)
 - i. Code vs. No Code
- g. [Solutions in Power Apps - Power Apps | Microsoft Learn](#)
 - i. Used to transport apps and components from one environment to another or to apply a set of customizations to existing apps.
 - ii. Can contain one or more apps as well as other components such as site maps, tables, processes, web resources, choices, flows, and more.
 - iii. For implementing application lifecycle management (ALM) in Power Apps and other Power Platform products, such as Power Automate.
6. Galleries provide a powerful way to visualize tabular data
 - a. Select the template by selecting in the first cell of the gallery, or you can select the pencil icon in the upper-left corner when the entire gallery is selected.
 - b. Test your gallery directly on the canvas by pressing the Alt key
7. Compare, you add it to a collection called CompareList.
8. Main Screen and Preview the app by selecting the Play button in the upper right.
9. Copresence within Power Apps Studio that allows you to collaborate with others in canvas apps.
10. [Power Apps pattern: Project management - Power Apps | Microsoft Learn](#)

a. Donations

11. Endlist

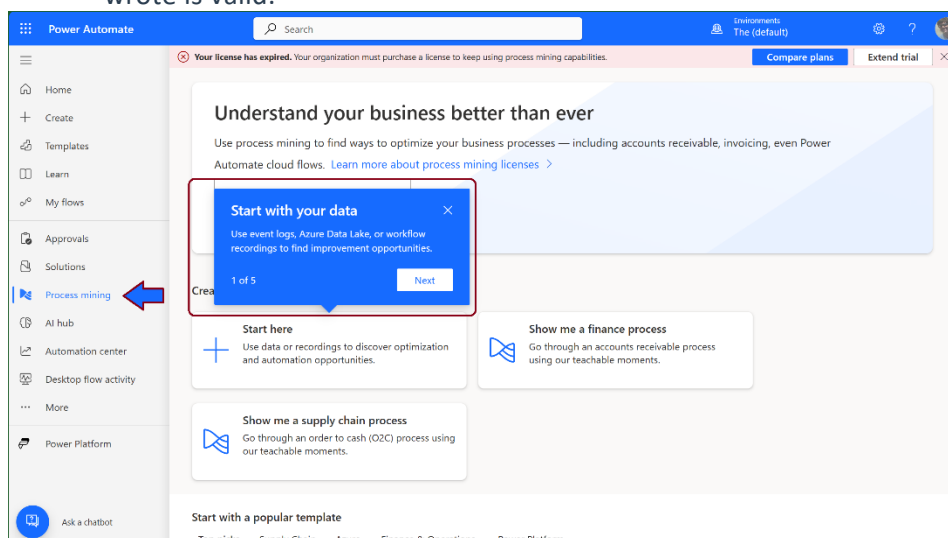
Updates 2025 – Roll Up the Sleeves - What & Where Can U - C.R.U.D.



Big Buckets

1. [Environment operations - Training | Microsoft Learn](#)
2. Sandbox, Production, Dev or Trial (subscription-based): History
3. Manage Business Units (BUs)
 - a. RBAC implement security boundaries by assigning roles and permissions at the BU level, restricting access to specific environments or resources.
4. Dashboards & Reports

5. Power Apps and Power Automate Tools
 - a. Embed Apps in Forms
6. Power Platform connectors
7. Dataverse Managed, Scalable data platform
8. Create your automation with Copilot
 - a. Process Mining Ingestion
 - i. Navigates you through the ingestion experience in Process Mining.
 - ii. Use event logs, Azure Data Lake, or workflow recordings
 - b. Process Mining Process Analytics helps you generate process insights through natural language.
 - c. Copilot can then take the data you collected and easily summarize findings from it quantitatively and qualitatively.
 - d. [Get started with Copilot in cloud flows - Power Automate | Microsoft Learn](#)
 - e. Toggle off the Cross-geo data sharing settings - Power Platform admin center
 - f. Copilot in Cloud Flows - create and edit Power Automate flows just by describing them in everyday language
 - g. Text generation model in AI Builder lets you use the GPT model directly in your Power Automate flows and your apps built in Power Apps
 - h. [Text generation model \(preview\) \(deprecated\) | Microsoft Learn](#)
 - i. [Use your prompt in Power Automate | Microsoft Learn](#)
 - j. [Quickstart - Getting started with Azure OpenAI Assistants \(Preview\) - Azure OpenAI | Microsoft Learn](#)
 - k. [Azure AI Foundry](#) portal & Azure OpenAI resource
 - l. The Assistants playground allows you to explore, prototype, and test AI Assistants without needing to run any code.
 - m. Confirm that the model used code interpreter to get to this answer and the code it wrote is valid.



- 9.
- 10.

What is Work or Task aka Dataverse Solution

11. In Order of Steps

- a. Business Process Flow (BPF)
- b. Workflows: Associate workflows with stages to automate specific actions at each stage.
- c. Workflows can send emails, update fields, or trigger further automation.
- d. Flow Steps: Within workflows, define individual steps performing specific actions.
- e. These can be manual tasks like user approvals or automated actions like data updates.

12. Humans or Automate Until Approval Step

- a. How

13. Source Connect Target

- a. Triggers
- b. Actions: Steps performed within the flow, like sending emails, creating records, or performing calculations.
- c. Conditions: Branch the flow based on specific criteria.
- e. Loops: Repeat actions a set number of times or until a condition is met.
- f. Variables: Store and reference information throughout the flow.
- g. Connectors: Bridge the gap between Power Automate and other services. Wash Rinse Repeat
- h. Schema: Definition of data exchanged between Power Automate and the service.
- i. Branches: Split the flow based on conditions using "If" statements.
- j. ? Loops: Repetitive actions using "For Each" or "Until" loops.
- k. ? Conditions: Evaluate criteria to determine flow direction.
- l. ? Error Handling: Capture and manage errors using "Try...Catch" blocks and continue execution.
- n. ? Variables: Store and manipulate data within the flow using dynamic values.

14. What is the Riverbed or Highway

15. What is the Baton being Passed in the Relay Race

16. [Explore Power Automate's automation center - Power Automate | Microsoft Learn](#)

- a. Comprehensive visualizations that enable you to monitor the health of your automations, quickly detect issues or trends and troubleshoot problems
- b. Underlying data is managed through Dataverse tables, secured via role-based access control (RBAC). In standard Dataverse environments (production, trial, sandbox, developer),

17. Workflow logs

- a. Viewing logs: Access workflow logs through the Power Platform admin center or directly
- b. within the specific workflow designer.
- c. Filtering and searching: Filter and search logs based on specific criteria like date, time,
- d. workflow name, or error code to locate relevant information quickly.
- e. Understanding log data: Interpret the information within the logs, which may include
- f. timestamps, user information, trigger events, execution details, and error messages.

- g. Alerting and reporting: Configure alerts to notify you of critical errors or unexpected behavior within the workflow. Additionally, use the logs to generate reports for auditing purposes or analyzing workflow performance trends.
- 18. Backup and restore environments:
 - a. Regularly back up environments to facilitate recovery in case of errors or accidental data loss.
 - b. BUP B4 Delete Resources?
- 19. [Auditing - Training | Microsoft Learn](#)
- 20. [Use Dataverse community tools for data manipulation - Training | Microsoft Learn](#)
 - a. XrmToolBox
- 21. Azure Synapse Link to Dataverse
 - a. Handle high-volume data scenarios with near real-time information.
 - b. Enables continuous replication of Dataverse table data to either an Azure Data Lake Storage Gen 2 account or an Azure Synapse Workspace. You can then use it to run analytics such as Power BI reporting, Microsoft Azure Machine Learning, Data Warehousing, and other integration scenarios.
- 22. [AI - Training | Microsoft Learn](#)
- 23. Cognitive Services is a suite of prebuilt AI services
- 24. [Use QR codes - Training | Microsoft Learn](#)
 - a. Barcode scanner control in Power Apps, you can use your mobile phone or device to scan a barcode
- 25. [Exercise - Embed your app into other Microsoft applications - Training | Microsoft Learn](#)
- 26. [Data format options | Databricks Documentation](#)
- 27. Endlist

Reference

1. [Describe Power Platform administration - Training | Microsoft Learn](#)
2. [Microsoft Copilot Studio | Extend Copilots or Create Your Own](#)
3. [Overview of building canvas apps - Power Apps | Microsoft Learn](#)
4. Copilot Studio is an end-to-end conversational AI platform that empowers you to create and customize copilots using natural language or a graphical interface.
5. Power Apps maker portal <https://make.powerapps.com>
6. Power Platform admin center <https://admin.powerplatform.microsoft.com>
7. [Power Platform CLI support for Power Pages | Microsoft Learn](#)
8. [General availability deployment - Power Platform | Microsoft Learn](#)
9. [About the Power Apps Developer Plan - Power Platform | Microsoft Learn](#)
 - a. Power Platform subscription sign up for the Power Apps Developer plan
10. [Temporal tables - SQL Server | Microsoft Learn](#)
11. [Introduction to NoSQL - GeeksforGeeks](#)
12. <https://learn.microsoft.com/en-us/entra/id-governance/entitlement-management-overview>
13. End List